

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	23-08-2026

Code of Conduct

I Aswanth N N (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Aswanth N N

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

Ans:

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.

Answer:

The sentence is:

“I reported the damage to the car after the accident.”

The ambiguity occurs because the prepositional phrase **“to the car”** can be attached to different parts of the sentence.

A simplified CFG can be represented as:

$S \rightarrow NP VP$

$VP \rightarrow V NP PP$

$NP \rightarrow Det N$

$PP \rightarrow P NP$

Interpretation 1: “to the car” modifies “reported”

S

```
|— NP → I
└— VP
    |— V → reported
    |— NP → the damage
    └— PP
        |— P → to
        └— NP → the car
```

This interpretation treats **“to the car”** as information related to the reporting action.

Meaning:

I reported the damage to the car.

Interpretation 2: “to the car” modifies “damage”

S

```
|— NP → I
└— VP
    |— V → reported
    └— NP
        |— NP → the damage
        └— PP
            |— P → to
            └— NP → the car
```

Here, **“to the car”** describes the damage.

Meaning:

I reported the damage that happened to the car.

The phrase **“after the accident”** can also be attached to the reporting event or the damage event, creating additional ambiguity.

b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.

Answer:

A basic CFG defines grammatical structures using production rules, but it has several limitations.

1. Ambiguity:

A CFG can generate multiple valid parse trees without deciding which one is correct.

2. No_probability:

A basic CFG does not assign probabilities to different interpretations. Therefore, it cannot automatically select the most likely meaning.

3. Agreement_problems:

Basic CFG rules do not naturally represent features such as singular/plural or subject-verb agreement.

4. Long_sentences:

Customer descriptions can contain many clauses and phrases, resulting in a large number of possible parse trees.

5. Conversational_language:

Customers may use incomplete sentences such as:

“Car damaged yesterday, need claim.”

A strict CFG may not recognize such input easily.

6. Context

limitations:

CFG mainly focuses on syntax and does not understand the real-world meaning of insurance claims.

Therefore, basic CFG alone is not sufficient for a reliable insurance claim-processing system.

c) Design an improved parsing approach using PCFG, Feature Structures, and Earley Parsing to resolve ambiguity and efficiently process the claim.

Answer:

The improved system can use the following architecture:

Customer Input



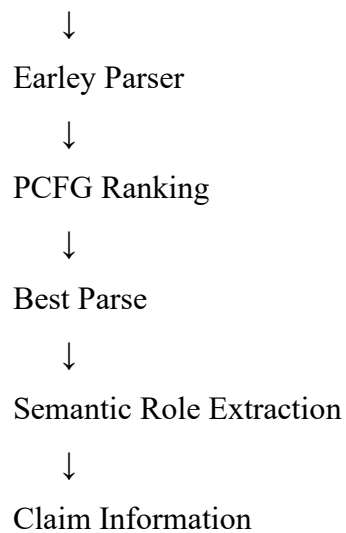
Tokenization



CFG Grammar



Feature Structures



PCFG

A **Probabilistic Context-Free Grammar** assigns probabilities to grammar rules.

If two possible structures exist, their probabilities can be compared and the more likely interpretation can be selected.

For example:

$VP \rightarrow V \ NP \ PP$

$NP \rightarrow NP \ PP$

can have different probabilities based on training data.

Feature Structures

Feature structures represent grammatical properties such as:

Number = Singular

Person = Third

Tense = Past

Voice = Active

They help enforce grammatical agreement.

Earley Parsing

Earley Parsing is a chart-based parsing algorithm that can efficiently handle:

- ambiguous grammars
- incomplete input
- different possible parse structures
- complex sentences

The parser can maintain alternative interpretations instead of immediately discarding them.

d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

Answer:

The proposed architecture improves the system in several ways.

Accuracy:

PCFG helps select the most probable interpretation from multiple possible parses.

Grammar

correctness:

Feature structures help enforce subject–verb agreement and other grammatical constraints.

Ambiguity

handling:

Earley Parsing can maintain multiple possible interpretations.

Scalability:

The system can process complex customer descriptions more effectively.

Reliability:

Semantic information can be validated before performing insurance operations.

For example, the system can convert the sentence into:

Claimant = Customer

Object = Car

Cause = Accident

Action = Report Damage

This structured information can then be passed to the insurance claim-processing system.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates "**with AC sleeper**" with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.

d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

Ans:

Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

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- Slow response for complex queries

Example passenger request:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

The system sometimes incorrectly associates “**with AC sleeper**” with the passengers instead of the ticket/class information.

Tasks:

a) Analyze the possible syntactic interpretations of the given passenger request using CFG.

Answer:

The request is:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

A simplified CFG is:

$S \rightarrow VP$

$VP \rightarrow V NP PP PP PP PP$

$NP \rightarrow \text{Num } N$

$PP \rightarrow P NP$

The intended structure is:

S

└─ VP

└─ V → Book

└─ NP → two tickets

└─ PP → from Chennai

|— PP → to Delhi
|— PP → for my parents
└— PP → with AC sleeper

The intended meaning is:

Number = 2

Source = Chennai

Destination = Delhi

Passengers = Parents

Class = AC Sleeper

However, another interpretation may attach:

“with AC sleeper”

to:

“my parents”

This could incorrectly imply that **AC sleeper** describes the passengers.

The correct interpretation is that **AC sleeper is a ticket/class attribute**.

b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.

Answer:

Top-down parsing starts from the start symbol and tries to construct the sentence using grammar rules.

For ambiguous input, it may select an incorrect grammar rule first. When that rule fails later, the parser must return to an earlier point and try another rule.

This process is called **backtracking**.

For example, the parser may first interpret:

with AC sleeper → passenger information

and later discover that this interpretation is incorrect.

It must then backtrack and try:

with AC sleeper → ticket class

For long conversational requests, many such possibilities can occur.

Therefore, excessive backtracking can cause:

- increased computation
- slower response
- repeated parsing
- poor real-time performance

This is undesirable for a railway booking system where users expect immediate responses.

c) Analyze how Earley Parsing can handle ambiguity, incomplete input, and different possible parse structures more effectively.

Answer:

Earley Parsing uses a chart to store partial parsing states.

It mainly uses three operations:

1. **Prediction**
2. **Scanning**
3. **Completion**

For example, when the user says:

“Book two tickets from Chennai...”

the parser can already build a partial structure.

When the user continues:

“to Delhi for my parents...”

the parser extends the existing structure.

When it reaches:

“with AC sleeper”

it can maintain different possible attachments and select the most appropriate interpretation later.

Therefore, Earley Parsing is useful for:

- ambiguous sentences
- partial input
- incomplete requests
- long-distance dependencies
- conversational language

d) Compare Top-Down Parsing, CFG-based parsing, and Earley Parsing in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

Answer:

Feature	Top-Down Parsing	CFG-Based Parsing	Earley Parsing
Grammar	CFG rules	CFG rules	CFG-compatible
Ambiguity	Poor handling	Can represent ambiguity	Handles ambiguity well
Backtracking	Often high	Depends on implementation	Reduced using chart

Partial input	Limited	Limited	Good
Long sentences	Can become inefficient	Can become complex	Better suited
Probabilities	Not inherent	Not inherent	Can combine with PCFG
Conversational input	Limited	Moderate	Good
Real-time suitability	Low to moderate	Moderate	High

For a railway booking assistant, **Earley Parsing combined with PCFG and Feature Structures** is a better choice because it can handle ambiguity and partial input while reducing unnecessary recomputation.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

Ans:

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved

- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

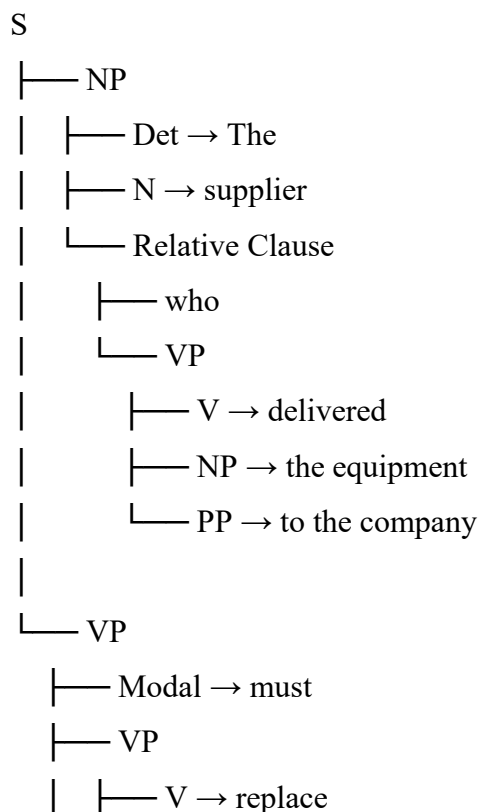
a) Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.

Answer:

The sentence is:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

A simplified structure is:



| └─ NP → the defective components
└─ PP → within thirty days

The relative clause:

“who delivered the equipment to the company”

describes **the supplier**.

Therefore:

Supplier → Agent

Equipment → Object

Company → Recipient

The main clause states:

supplier must replace defective components.

Possible ambiguity can arise from the attachment of prepositional phrases and the relationship between the relative clause and the main noun.

b) Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.

Answer:

Basic CFG can represent grammatical structures, but it does not automatically understand their semantic relationships.

Legal documents often contain:

- relative clauses
- nested clauses
- passive constructions
- multiple prepositional phrases
- long-distance dependencies
- complex noun phrases
- conditions
- exceptions

For example:

“The supplier who delivered the equipment to the company...”

The CFG can identify the relative clause, but it does not automatically know that **who** refers to the supplier.

Similarly, CFG does not inherently determine that:

supplier = delivery agent

and:

company = recipient.

Therefore, syntax alone is insufficient for legal semantic interpretation.

c) Design an NLP architecture integrating CFG, PCFG, Feature Structures, and Earley Parsing to improve syntactic and semantic interpretation.

Answer:

The proposed architecture is:

Legal Document



Sentence Segmentation



Tokenization



CFG Grammar



Feature Structures



Earley Parsing



PCFG Ranking



Semantic Role Labeling



Entity Extraction



Obligation Detection



Legal Knowledge Representation

CFG

Defines possible grammatical structures.

PCFG

Assigns probabilities to alternative parse trees and selects the most probable structure.

Feature Structures

Store information such as:

Number

Person

Tense

Voice

Agreement

They help maintain grammatical consistency.

Earley Parsing

Handles ambiguity, recursive structures, incomplete input, and complex sentences efficiently.

Semantic Role Labeling

Maps syntactic structures to semantic roles such as:

Agent

Object

Recipient

Source

This allows the system to understand the actual meaning of the contract.

d) Develop a step-by-step processing workflow showing how the system can extract:

- Supplier
- Action
- Equipment
- Company
- Obligation
- Deadline

and justify how the proposed approach improves automated contract analysis.

Answer:

Step 1: Tokenization

The sentence is divided into tokens:

The | supplier | who | delivered | the | equipment |
to | the | company | must | replace | the |
defective | components | within | thirty | days

Step 2: CFG Parsing

The parser identifies:

- main clause
- noun phrase
- relative clause
- verb phrases
- prepositional phrases

Step 3: Relative clause analysis

The system identifies:
who delivered the equipment to the company
as a relative clause modifying **supplier**.

Therefore:

Supplier = Delivery Agent

Equipment = Delivered Object

Company = Recipient

Step 4: Main action extraction

The main action is:

replace

The system identifies:

Supplier → must replace → defective components

Step 5: Obligation extraction

The word:

must

indicates an obligation.

Therefore:

Obligation = Supplier must replace defective components

Step 6: Deadline extraction

The phrase:

within thirty days

is identified as the deadline.

Deadline = 30 days

Final structured output

Supplier = Supplier

Action = Replace

Equipment = Equipment

Company = Recipient

Obligation = Replace defective components

Deadline = Within 30 days

This approach improves legal contract analysis because it combines syntactic structure, probability, grammatical features, and semantic relationships instead of relying only on basic CFG rules.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- c) Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

Ans:

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.

Answer:

The query is:

"Find laptops with a touchscreen for students under ₹60,000."

The intended interpretation is:

Product = Laptop

Feature = Touchscreen

Target User = Students

Price = Under ₹60,000

A simplified CFG structure is:

S

└─ VP

└─ V → Find

└─ NP

└─ N → laptops

└─ PP → with a touchscreen

└─ PP → for students

└─ PP → under ₹60,000

The intended interpretation is:

Find laptops that have a touchscreen, are suitable for students, and cost under ₹60,000.

Another possible interpretation could attach:

“for students”

to:

“touchscreen”

This could incorrectly imply that the touchscreen itself is specifically for students.

Semantic analysis is therefore required to choose the correct interpretation.

b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.

Answer:

Basic CFG has several limitations.

Ambiguous attachment

The same phrase may attach to different parts of the sentence.

Informal queries

Users may type:

“good laptop for college under 60k”

instead of a complete grammatical sentence.

Elliptical queries

Users may omit important words:

“Laptop touchscreen under 60k.”

The system must infer the intended meaning.

Missing words

The system may need to infer:

Find laptops...

even when the user only types:

laptops under 60k.

Long queries

Long product descriptions can create many possible parse trees.

Therefore, CFG alone is not sufficient for modern conversational product search.

c) Propose an improved parsing architecture using PCFG, Feature Structures, and Earley Parsing to identify product attributes, user requirements, and price constraints.

Answer:

The proposed architecture is:

User Query



Tokenization



CFG Parsing



Feature Structures



Earley Parsing



PCFG Ranking



Semantic Attribute Extraction



Structured Search Query



Product Retrieval

Feature structures can represent:

Product = Laptop

Feature = Touchscreen

TargetUser = Student

MaximumPrice = ₹60,000

PCFG can assign probabilities to different interpretations.

Earley Parsing can maintain alternative parses and handle incomplete queries.
Semantic analysis then converts the selected parse into product-search attributes.

d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

Answer:

Search precision

The system correctly identifies:

Product = Laptop

Feature = Touchscreen

Target = Students

Price $\leq$ ₹60,000

This allows the search engine to return relevant products.

Response time

Earley Parsing can efficiently process complex and partial input, while structured output allows the search engine to directly apply filters.

Customer experience

Customers can use natural language instead of manually selecting:

- Product
- Feature
- Price
- Target user

The system therefore provides a more natural conversational shopping experience.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "**with two passengers**" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.
- b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Ans:

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
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Example spoken command:

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Depending on the parse, **“with two passengers”** may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

“Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

Tasks:

a) Analyze the possible syntactic structures and ambiguities in the given voice command.

Answer:

The command is:

“Change my flight to Mumbai tomorrow with two passengers.”

The intended structure is:

Action = Change

Flight = Existing flight

Destination = Mumbai

Date = Tomorrow

Passengers = 2

A simplified structure is:

VP

└─ V → Change

└─ NP → my flight

|— PP → to Mumbai

|— AdvP → tomorrow

|— PP → with two passengers

The ambiguity occurs with:

“with two passengers”

The phrase could be incorrectly attached to the destination:

Mumbai with two passengers

However, semantically it represents the number of passengers associated with the booking.

Therefore:

Passengers = 2

not:

Destination = Mumbai with two passengers

The correction:

“actually, make that Bangalore”

creates another interpretation issue. The system must understand that **Bangalore replaces Mumbai**.

b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.

Answer:

Top-down parsing starts from the start symbol and attempts to construct a complete parse.

Spoken commands create additional challenges because they can be:

- incomplete
- interrupted
- corrected
- ambiguous
- affected by speech-recognition errors

For example:

“Change my flight to Mumbai tomorrow...”

The parser may already select **Mumbai** as the destination.

The passenger then says:

“actually, make that Bangalore.”

The system must revise its previous interpretation.

A conventional top-down parser may require backtracking and reparsing, increasing response time.

Therefore, it is less suitable for real-time voice interaction.

c) Design a parsing architecture using Earley Parsing, PCFG, and Feature Structures to support partial input, ambiguity resolution, and agreement constraints.

Answer:

The proposed architecture is:

Voice Input



Speech Recognition



Tokenization



Earley Parsing



Feature Structures



PCFG Ranking



Semantic Interpretation



Correction Detection



Booking Validation



Airline Booking System

Earley Parsing

Allows partial input to be processed without requiring the entire sentence to be complete.

PCFG

Ranks competing interpretations using probabilities.

For example:

with two passengers

can be assigned a higher probability as a passenger constraint rather than a destination attribute.

Feature Structures

Maintain structured information:

Action = Change

Destination = Mumbai

Date = Tomorrow

Passengers = 2

When the passenger says:

“Actually, make that Bangalore.”

the feature structure is updated:

Destination = Bangalore

The old value **Mumbai** is replaced.

d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Answer:

Step 1: Speech recognition

The passenger's voice is converted into text:

“Change my flight to Mumbai tomorrow with two passengers.”

Step 2: Tokenization

Change

my

flight

to

Mumbai

tomorrow

with

two

passengers

Step 3: Earley Parsing

The system generates possible syntactic structures and handles ambiguous attachments.

Step 4: PCFG ranking

The most probable interpretation is selected.

Step 5: Feature extraction

The system creates:

Action = Change

Flight = Existing Booking

Destination = Mumbai

Date = Tomorrow

Passengers = 2

Step 6: Correction detection

If the user says:

“Actually, make that Bangalore.”

the system detects a correction and updates:

Destination = Bangalore

Step 7: Validation

The system verifies:

- destination
- travel date
- passenger count
- flight availability
- booking details

Step 8: Booking modification

Only after validation does the system send the structured request to the airline booking system.

Final structured request

Action = Change Flight

Destination = Bangalore

Date = Tomorrow

Passengers = 2

The proposed method is suitable for real-time airline applications because **Earley Parsing handles partial and ambiguous input**, **PCFG helps select the most probable interpretation**, and **Feature Structures maintain consistent booking information and support corrections**.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	4	4	4	3	3	3	21	94
2	4	4	4	3	3	3	21	
3	3	4	4	4	3	3	21	
4	4	4	4	3	3	3	21	
5	4	4	4	3	3	3	21	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____